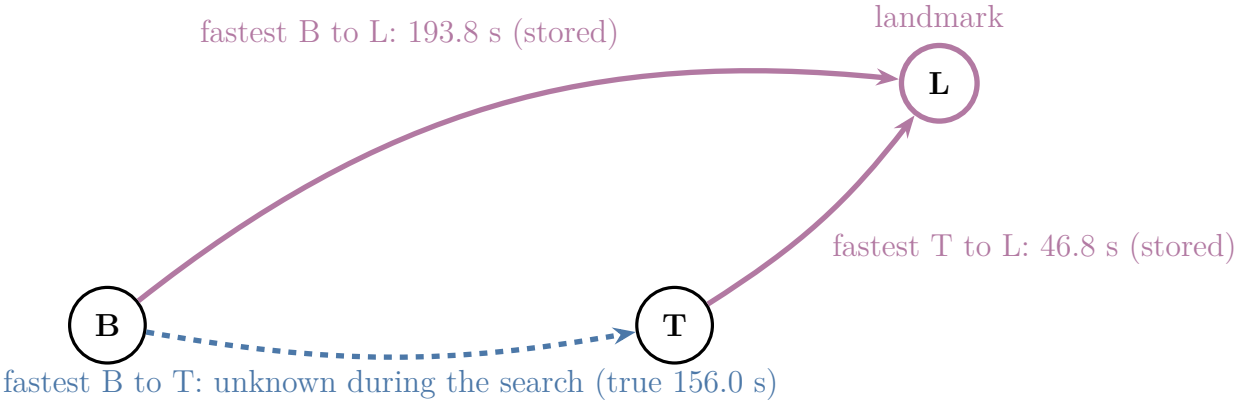

$B \to T \to L$ is one way to reach L, never faster than the fastest:

$$\text{time}(B, T) + 46.8 \geq 193.8$$

$$\text{time}(B, T) \geq 193.8 - 46.8 = 147.0 \text{ s}$$